

Accounting Software - Requirement & Module Flow

This document explains the complete flow and module structure of the Advanced Accounting Software project. The system will be developed using React JS, Node JS, TypeScript, PostgreSQL and Tailwind CSS.

Overall System Flow

- Admin creates company setup.
- Users login based on their role.
- Products, customers and suppliers are managed.
- Purchases increase inventory stock.
- Sales invoices reduce stock automatically.
- Accounting entries are created automatically.
- GST and tax reports are generated.
- Dashboard and analytics show business performance.
- Reports can be exported to PDF and Excel.
- Audit logs and backups maintain security and data safety.

1. Authentication & User Management

Features:

- Login / Logout
- JWT Authentication
- Role Based Access (Admin, Accountant, Staff, Auditor)
- Change Password
- User Activity Logs
- Session Management

Relations: Connected with every module for security and permissions.

2. Company Setup Module

Features:

- Company Profile
- GST Number
- Financial Year Setup
- Bank Details
- Invoice Settings
- Multi Company Support

Relations: Used by Invoice, GST, Reports and Accounting modules.

3. Dashboard & Analytics

Features:

- Sales Overview
- Purchase Overview
- Profit & Loss Summary
- Low Stock Alerts
- Pending Payments
- Graphs & Charts

Relations: Fetches data from Sales, Purchase, Inventory and Accounting.

4. Customer Management

Features:

- Customer CRUD
- GST Details
- Credit Limit
- Customer Ledger
- Payment History
- Search & Filters

Relations: Connected with Sales, Payments and Reports.

5. Supplier / Vendor Management

Features:

- Supplier CRUD
- Purchase History
- Outstanding Payments
- Vendor Ledger

Relations: Connected with Purchase and Accounting modules.

6. Product & Inventory Module

Features:

- Product CRUD
- Barcode / SKU
- Categories & Units
- Warehouse Management
- Stock Tracking
- Low Stock Alerts
- Expiry Tracking

Relations: Connected with Purchase, Sales and Reports.

7. Purchase Module

Features:

- Purchase Invoice
- Supplier Selection
- GST Calculation
- Purchase Return
- Bulk Product Entry

Relations: Increases stock and creates supplier payable entries.

8. Sales & Billing Module

Features:

- GST Invoice
- POS Billing
- Discount Management
- Invoice PDF
- Email / WhatsApp Invoice
- Sales Return

Relations: Reduces stock and creates customer receivable entries.

9. Accounting Module

Features:

- Journal Entries
- Ledger Management
- Profit & Loss
- Balance Sheet

Relations: Connected with Sales, Purchase, Expenses and Payments.

10. Expense Management

Features:

- Daily Expenses
- Expense Categories
- Receipt Upload
- Expense Reports

Relations: Used in Profit & Loss calculations.

11. Payment Module

Features:

- Partial Payments
- UPI / Cash / Card / Bank
- Pending Due Tracking
- Payment Reminders

Relations: Connected with Customer, Supplier and Accounting modules.

12. GST & Tax Module

Features:

- GST Reports
- Tax Calculation
- HSN/SAC Support
- Tax Summary
- GST Export Files

Relations: Connected with Sales, Purchase and Accounting.

13. Payroll Module

Features:

- Employee Salary
- Bonus & Deductions
- Attendance Based Payroll
- Salary Slips

Relations: Connected with Employee and Expense modules.

14. Reports Module

Features:

- Sales Reports
- Purchase Reports
- Inventory Reports
- Tax Reports
- Customer & Supplier Ledger
- Excel / PDF Export

Relations: Collects data from all major modules.

15. Notification & Reminder Module

Features:

- Due Payment Alerts
- Low Stock Notifications
- Invoice Reminders
- Email Notifications

Relations: Connected with Inventory, Payments and Sales.

16. Audit Log & Backup Module

Features:

- Track User Actions
- Data Backup
- Restore Backup
- System Activity History

Relations: Connected with all modules for monitoring and security.

This project will follow a scalable and modular architecture with clean code, reusable components, role-based security, advanced reporting and professional UI/UX suitable for production use.